all samples were performed. Three different isolates were obtained from all samples. On the basis of their biochemical study and sugar test, the isolates were identified as *Lactobacillus casei*, *Lactobacillus brevis* and *Lactobacillus fermentum*. The antagonistic study of these *Lactobacilli* species were performed against four pathogenic bacteria- *Pseudomonas aeruginosa, Escherichia coli, Staphylococcus aureus* and *Proteus* sp.. The antagonistic study was performed by well diffusion method. The inhibition zone against *Pseudomonas aeruginosa, Escherichia coli* and *Proteus sp.* was observed maximum by *Lactobacillus brevis* and against *Staphylococcus aureus*, the maximum inhibition zone was observed by *Lactobacillus fermentum*.

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Habenaria gibsonii var. *foetida* Blatt. & McCann (Orchidaceae): New distribution recorded from Sitamata Wildlife Sanctuary, Rajasthan, India

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Abstract- Orchid species *Habenaria gibsonii* var. *foetida* Blatt. & McCann (Orchidaceae) is reported from Sitamata Wildlife Sanctuary of Rajasthan, India for the first time along with its description, illustration, photographs and location in Rajasthan.

Key words: *Habenaria gibsonii* var. *foetida*, Orchidaceae, Sitamata Wildlife Sanctuary, Description, Report

INTRODUCTION

During the present investigation and characterization of orchid plants of Rajasthan, an interesting Orchid species was collected from Sitamata Wildlife Sanctuary, Rajasthan. After critical studies of the collected sample, this orchid was identified as *Habenaria gibsonii* var. *foetida*. Almost all floristic research work has been examined critically but *Habenaria gibsonii* var. *foetida* Blatt. & McCann has not been reported from Sitamata Wildlife Sanctuary, Rajasthan to date.[1-15]

Description:

Habenaria gibsonii var. *foetida* Blatt. & McCann, J. Bombay Nat. Hist. Soc. 16 (1932); Tiwari & Maheshwari, Orch. Madh. Pra.: 435, (1963); Misra, Orch. Ori.: 186 (2004).

Terrestrial, 30-40 cm high, Tuber single, spheroid or oblong ellipsoid, vertically below the stem roots few. Stem cylindrical. Leaves 5-8, dark green above and pale below, alternate, sessile, elliptic-lanceolate, acute, slightly undulate, base narrow, sheathing below. Inflorescence erect, 14-20 cm long, peduncle cylindrical, with large obovate lanceolate, acuminate, sub-sheathing sterile bracts subtending raceme. Raceme lax, not compact, up to 10-12 flowers. Bracts longer than ovary, upper ones smaller, ovate-lanceolate, subulate, 5-veined, green. Flower foul scented, greenish-white. Pedicel 1-2 cm long. Sepals unequal, concave, dorsal erect, hooded, broadly ovate, sub-obtuse; laterals longer, spreading to the base, oblong-lanceolate, rounded, falcate. Petals bipartite nearly to the base, segments linear, tapering upward, acute, upper strongly sickle-shaped, lower segment almost at a right angle to the upper segment, strongly incurved upward. Lip 3-partite nearly to the base with subequal segments, midsegment decurved; laterals forming acute angles with the middle, reflexed, decurved upwards. Spur shorter and close to the ovary, clavate with an obtuse apex. Column erect, more or less oblong. Stigmatic processes stretched forward, slightly divergent, linear narrowing the apices.

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Rostellum broad and triangular, rounded. Anther locules waxy yellow, parallel, broad. Pollinia obliquely obovate, apex bilobed. Ovary 1-2 cm long, slightly curved at the apex, ribbed, green.

Flowering: August, **Fruiting**: September – October.

Ecological notes: This species grows underneath of mixed type forest with tree species like *Terminalia tomentosa, Madhuca indica, Mangifera indica, Terminalia bellirica, Anogeissus pendula, Ficus benghalensis* and *Butea monosperma* etc. in humus-rich soil. The plant grows with herbaceous plants like *Desmodium* sp., *Diocscorea* sp., *Chlorophytum* sp. and other associated herbs.

Diagnostic features: *H. gibsonii* var. *foetida* can be differentiated in the field on the basis of foetid smelling flowers.

Distribution in India: Jharkhand, Madhya Pradesh, Chhattisgarh, Orissa, Maharashtra, Gujarat and Rajasthan

The Genus *Habenaria* is a large genus with around 600 species widely distributed all over the world. Within the Rajasthan, only six species are reported *vis. Habenaria digitata, Habenaria frucifera, Habenaria longicorniculata, Habenaria marginata, Habenaria plantaginea* and *Habenaria gibsonii.* Earlier *H. gibsonii* var. *foetida* Blatt. & McCann has been reported from Jharkhand, Madhya Pradesh, Chhattisgarh, Orissa, Maharashtra, Gujarat and Rajasthan (Sirohi District)[15-17]. This species is common in the Sitamata Wildlife Sanctuary and can be easily traced out during the flowering time. The previous record of this plant was only from Mount Abu Wildlife Sanctuary of Sirohi District. This is the first time it has been discovered in the Sitamata Wildlife Sanctuary, Rajasthan[15].





Fig. 1 *Habenaria gibsonii* var. *foetida* Blatt. & McCann: A. Whole plant in habitat; B. Plant Inflorescence

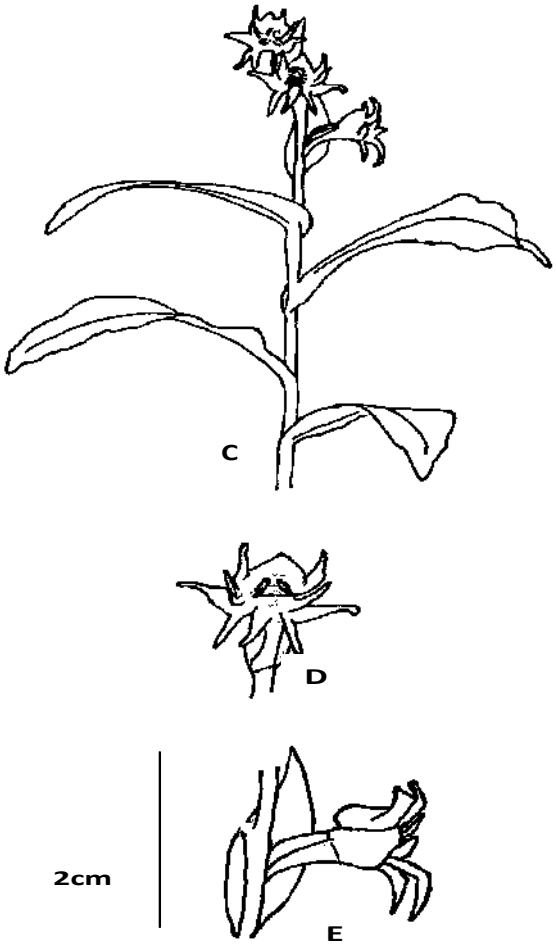


Fig. 2: *Habenaria gibsonii* var. *foetida* Blatt. & McCann: Illustration: C. Plant with inflorescence D. Close up of single flower E. lateral view of flower A